

Gyro-Enhanced Micromouse — Technical Report

Emmanuel Basua — Electrical & Computer Engineering

Overview

An autonomous maze-solving robot using time-of-flight sensing, wheel encoders and an IMU gyroscope to improve orientation estimation and closed-loop trajectory following.

System architecture

The platform combines a custom embedded controller, sensor acquisition, state estimation and motor control. Gyroscope bias compensation is used to reduce accumulated heading error.

Control approach

The controller combines measured heading and distance information to maintain stable wall following and repeatable turns through a maze.

Engineering outcome

The project demonstrates how sensor fusion and careful embedded implementation can improve autonomous navigation under practical hardware constraints.